COMPUTER SCIENCE AND ENGINEERING
(IOT, Cybersecurity Including Blockchain Technology)

YENEPOYA
INSTITUTE OF TECHNOLOGY

Iot Based Smart Waste Management System

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Problem Statement

Urbanization generates increased waste volume, stressing outdated collection systems. Bin overflow leads to health hazards, pollution, and traffic issues. Static truck routes cause resource waste and service inequity; current systems lack responsive, real-time monitoring. A smart system using IoT can provide real-time bin status, optimize collection routes, reduce unnecessary trips, and improve urban cleanliness and resource use.

Proposed System

Smart dustbins use sensors to detect when they are full and send this information online to a central system. When a bin is ready to be emptied, the system quickly finds the closest garbage truck and sends an alert to the driver, helping them collect waste efficiently.

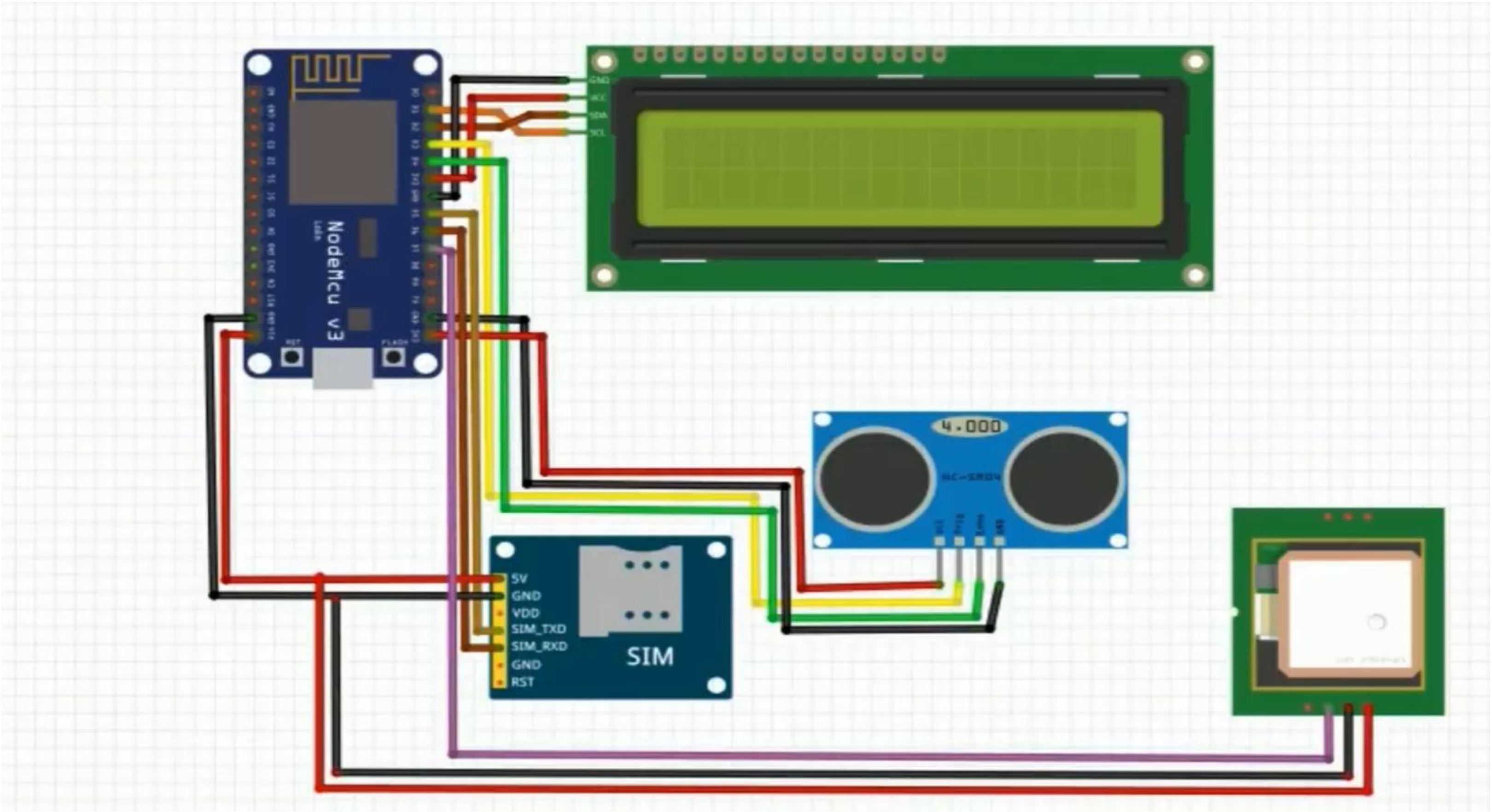
Key Features:

- Real-time bin monitoring
- Automatic alerts for full bins
- GPS-guided truck routing

Benefits:

- Cleaner streets and less overflowing bins
- Lower fuel costs for trucks
- Faster and smarter waste collection

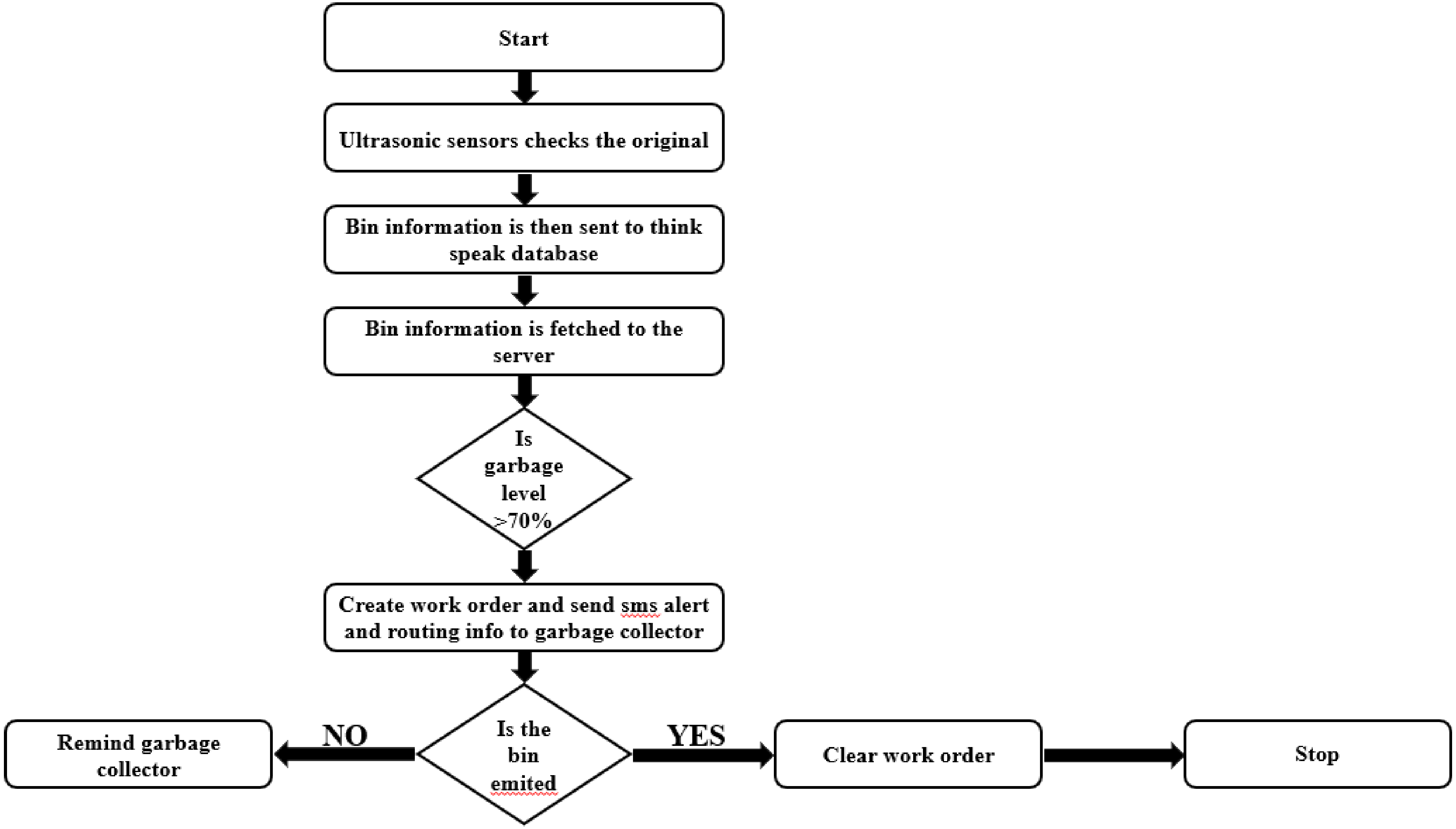
System Architecture



Module Description

Module	What it Does	How it Works with Others
Ultrasonic Sensor	Measures how full the bin is (empty, half, full)	Sends bin status to Microcontroller
Microcontroller	Controls the system and processes data	Gets data from sensors and sends commands
LCD Display	Shows bin status and system info	Gets info from Microcontroller
GPS Module	Tracks bin or truck location for route planning	Sends location to Microcontroller
GSM Module	Sends alerts and bin status to server or phone	Controlled by Microcontroller
Battery Unit	Powers the whole system	Powers all modules

Flow Diagram



Tools Used

- Programming Language: C/C++ (For Arduino) , Python (cloud processing)
- Platform: Arduino (Main board for sensors and control)
- Libraries: Basic Arduino sensor and display libraries
- Database: Cloud storage (For saving bin data)
- APIs: GSM (For sending messages), MQTT/HTTP (For data)
- Development Tools: Arduino IDE (For coding); Git

Challenges and Risk Faced

- Sensors and batteries may fail.
- Weak internet signals can disrupt data transfer.
- Devices need regular cleaning and maintenance.
- Protecting data privacy is crucial.

- Regular checks and maintenance.
- Use backup networks.
- Encrypt data for safety.

Future Scope

- Add AI to predict waste levels and optimize collection.
- Use solar power for bins to save energy.
- Include waste sorting robots for better recycling.
- Expand to cover larger city areas easily.
- Improve mobile apps for real-time monitoring and alerts.
- Integrate with other smart city services for efficiency.

Conclusion

This project designs an IoT-based smart waste management system to monitor bin levels and optimize garbage collection routes. The system uses sensors, microcontrollers, and cloud communication for real-time data and alerts.

References

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THANK YOU